

# Designing for Human Responses:

## Behavioral Insights for AI-Driven Retail Decisions

Framing · Reference · Context · Fairness · Green

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Executive MBA Session

# About the Presenter



## Mehmet Gumus

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Management  
Desautels Chair in Supply  
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Business Analytics

McGill University

### Academic background

Ph.D. in Industrial Engineering & Operations Research and  
M.A. in Economics, **UC Berkeley**; at McGill since **2007**.

### Research

Supply chain management, **dynamic pricing**, and risk  
management. Published in Management Science, Operations  
Research, MSOM, Marketing Science, and POM; Associate  
Editor for **POM** and **IIE Transactions**.

### Program leadership

Developed McGill's **Master of Management in Analytics (MMA)**  
in 2017 and its online version in 2023; Academic Director since  
its launch.

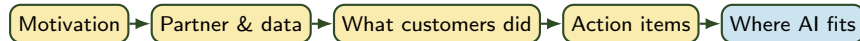
### Entrepreneurship

Founded **Logiciel Plannica Inc.** (2015): supply chain analytics  
that integrates with databases and ERP systems.

# Agenda

|   | Topic                                                                  | Setting / industry partner                   |
|---|------------------------------------------------------------------------|----------------------------------------------|
| ★ | <b>Why It Matters</b> <i>Human responses and AI-driven decisions</i>   | <i>Five lenses on customer response</i>      |
| 1 | <b>Framing effects</b> <i>Shipping fees or shipping free?</i>          | <i>Online retailers (BizRate price data)</i> |
| 2 | <b>Reference effects</b> <i>Raising free-shipping thresholds</i>       | <i>North American online grocer</i>          |
| 3 | <b>Context effects</b> <i>Is “eye level” really “buy level”?</i>       | <i>North American C-store chain</i>          |
| 4 | <b>Fairness</b> <i>Size-inclusive fashion assortments</i>              | <i>European online fashion retailer</i>      |
| 5 | <b>Green considerations</b> <i>Profit vs. environmental impact</i>     | <i>Choice experiment, UK shoppers</i>        |
| ★ | <b>Synthesis</b> <i>Cross-cutting lessons, AI playbook, discussion</i> | <i>Action list</i>                           |

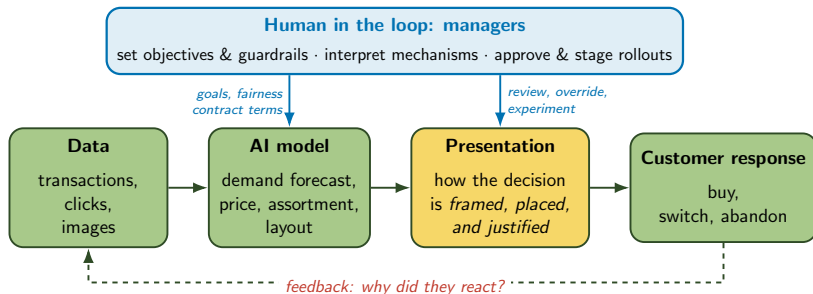
## Inside each of the five topics



# The Papers Behind This Session

- ❶ **Framing Effects:** Gumus, Li, Oh, Ray. *Shipping Fees or Shipping Free? A Tale of Two Price Partitioning Strategies in Online Retailing*. Production and Operations Management 22(4):758–776, 2013. [[https://mehmet-gumus.github.io/papers/J-POM\\_1.pdf](https://mehmet-gumus.github.io/papers/J-POM_1.pdf)]
- ❷ **Reference Effects:** Zhou, Jin, Gumus, Animesh. *How to (and How Not to) Increase Free-Shipping Thresholds: Evidence from Online Retail*. Working paper. [<https://mehmet-gumus.github.io/papers/J-Working-Threshold-Economics.pdf>]
- ❸ **Context Effects:** Jalali, Cohen, Ertekin, Gumus. *Revisiting the Eye-Level Effect*. Working paper. [<https://mehmet-gumus.github.io/papers/J-Working-Eye-Level.pdf>]
- ❹ **Fairness Considerations:** Alavi, Gumus, Nasiry. *Designing Size Inclusive Fashion Assortments*. Working paper. [<https://mehmet-gumus.github.io/papers/J-Working-Fairness.pdf>]
- ❺ **Green Consideration:** Matte, Gumus, Roederderk. *Green Synergy: A Three-Party Fashion Retail System Balancing Profit Maximization and Environmental Impact Minimization*. Working paper. [<https://mehmet-gumus.github.io/papers/J-Working-Green-Energy.pdf>]

# AI Optimizes What It Models — Customers Respond to What They See

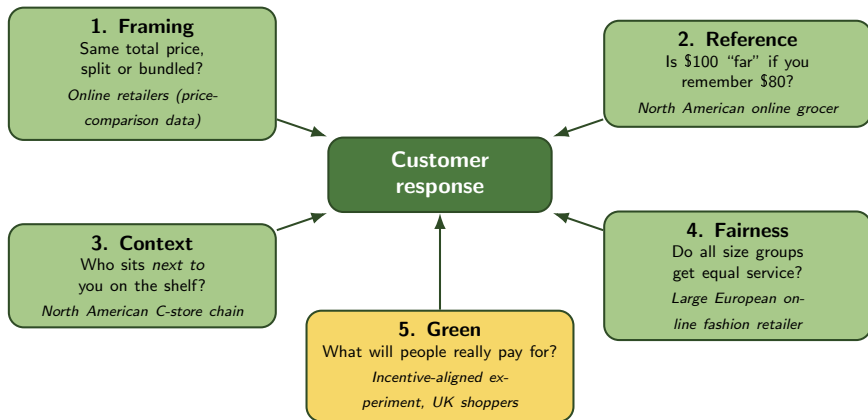


- Same price, product, or shelf — **different sales** depending on presentation. An AI that ignores this **mis-predicts** and **mis-prescribes**.
- Humans supply what data cannot: **objectives**, **guardrails**, **behavioral explanations**, and **staged rollouts**.

## Synthesis between AI and Human

The biggest wins come from AI that models **human responses** — steered by **humans in the loop** who set goals, test, and interpret.

# Five Lenses on Customer Response



# 1. Framing: “Shipping Fees or Shipping Free?”

## Why we studied it

- Shipping & handling (S&H) fees are consistently the **number-one complaint** about online shopping.
- Free shipping is expensive: Amazon spent **\$102.7B** on shipping in 2025 ( $\approx 14\%$  of net sales), up from \$95.8B in 2024.
- **Zero-price effect**: moving a fee from \$1 to \$0 lifts demand far more than \$2 to \$1.

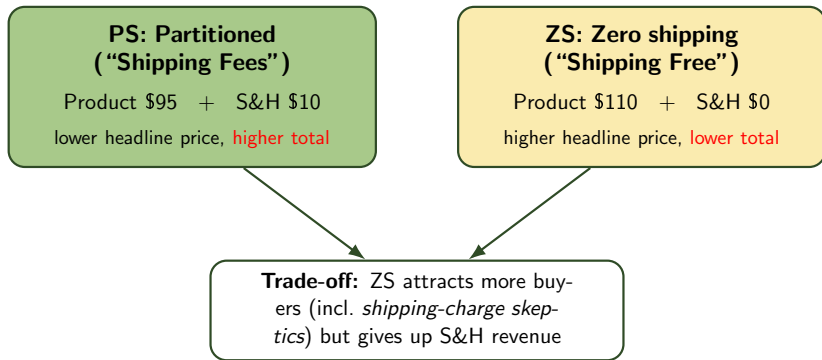
## A famous accident

When Amazon rolled out free shipping in Europe, France was mistakenly charged **1 franc** ( $\approx 10$  cents). Orders jumped everywhere *except* France — until the fee became exactly zero.

## Managerial question

Should an online retailer **partition** its price (product + S&H fee) or **bundle** it into one “free shipping” price — and which retailers win with each?

# Two Frames for the Same Economics

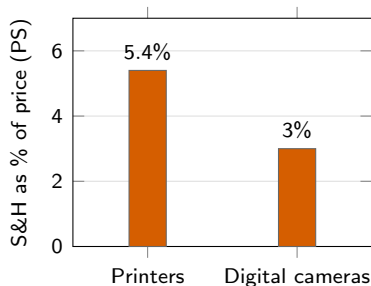


(Illustrative numbers.)

**Data:** A custom AI bot collected daily prices from **Online Platforms** for 4 months — **24 products** (digital cameras vs. printers), large and small retailers (e.g., Amazon, Dell, Buy.com).

# What We Found

Bulky products carry bigger surcharges



- Free-shipping retailers change prices  $\approx 1.5\times$  more often.
- $\approx 40\%$  of retailers already mix policies across products and time.
- Retailers selling large/heavy items (high S&H-to-price ratio) charge fees.

## Amazon's path fits the model

No free shipping 1995–1999 → free above \$100 → \$25 (2002) — only after scale and logistics efficiency made it affordable. Webvan went “free” too early and lost money on most items. Today: non-Prime free shipping above \$35 (raised from \$25 in 2023); Prime (\$139/yr) has no minimum.

# Framing: Action Items

## Offer free shipping when. . .

- 1 your S&H is **operationally efficient**;
- 2 you are **large/popular** enough to negotiate carrier rates;
- 3 many customers are **shipping-charge skeptics**;

## Charge a fee when. . .

- 1 products are **bulky/heavy**: customers expect to pay, and the subsidy would erode margin.
- 2 future costs (fuel, commodities) look **volatile**.

## AI Where AI fits

- Treat the **price frame** as a decision variable, not just the price level.
- Use the **S&H-to-price ratio** per SKU as a feature to decide which items are “shipping-included.”
- Guard dynamic pricing: frequent changes invite customers to **wait for cuts**.

## 2. Reference: Raising the Free-Shipping Threshold

### Why we studied it

- $\approx 65\%$  of online retailers use *contingent free shipping* (free above \$X).
- Rising fulfillment costs: the average threshold went from  $\approx \$52$  (2019) to  $\approx \$103$  (2025).
- Industry is guessing — and reversing:

### Costly U-turns

**Amazon Fresh:** \$35  $\rightarrow$  \$150  $\rightarrow$  \$100 within 10 months.

**Freshippo (Alibaba):** 49  $\rightarrow$  99 RMB, reverted within 2 months.

### Managerial question

Free-shipping thresholds make people **top up** their carts. Does that still work when you **raise** the threshold?

# Industry Partner and Natural Experiment

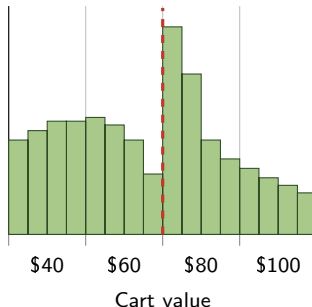
## Our partner

North American online grocery platform (founded 2019) connecting **1,000+ farmers and food processors** with consumers; **11,000 products** across  $\approx 400$  subcategories.

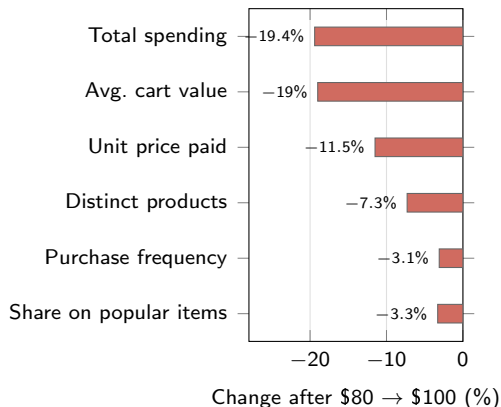
## The policy change

Metro customers: free shipping above **\$80**, raised to **\$100** in Nov 2023 — *no announcement*. Other regions: paid shipping, unchanged  $\Rightarrow$  **difference-in-differences**.

Shopping carts bunch just above \$80



# Raising the Threshold Backfired



- Customers **stopped topping up**: smaller, cheaper, more utilitarian baskets.
- Hit hardest: customers with **nearby grocery alternatives** and **smaller households**.
- The higher threshold did **not** solve the fulfillment-cost problem — it may have **reduced profit**.

# Why? Two Behavioral Mechanisms

## Reference point

Recent shoppers remember \$80, so \$100 feels like +25% — the goal feels unattainable.

Recent shoppers:  $\approx 55\%$   
larger drop in spending

## Search effort

Closing a \$20 larger gap means finding more items worth buying. Customers who slow down at the end of the cart give up.

High-effort shoppers:  
 $\approx 54\%$  larger drop

- Customers who had **not** shopped recently (old threshold forgotten) did **not** reduce spending.
- Mechanisms were measured from **clickstream**: purchase recency and time spent choosing the last items.

# Reference: Action Items

## Design rules

- 1 Raise the threshold only if the average cart is **close to** the current one (optimal policy is “all-or-nothing”).
- 2 **Roll out gradually**: show the new threshold only to customers who haven't bought in **2–3+ months**.

## Trade-off of the gradual rollout

Recovers most lost profit but slows adoption: 5-month cutoff  $\Rightarrow$  only **56%** see the new threshold after a year.

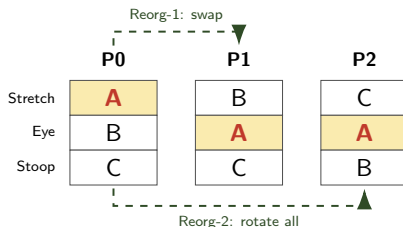
## AI Where AI fits

- Predict each customer's **reference point** (recency) and **search effort** from clickstream.
- Recommenders that **cut search effort**: “\$12 to go — your usual items.”

### 3. Context: Revisiting “Eye Level Is Buy Level”

#### Why we studied it

- Eye-level space is **scarce and monetized**: slotting fees exceed **16%** of new-product launch costs.
- Trade allowances exceed **\$50B per year**.
- Contracts promise “eye level” — but moving one product up forces **others to move**.



#### Managerial question

Product A reaches eye level in **both** P1 and P2. Does it matter **how** the rest of the shelf is rearranged?

# Where Is “Eye Level”? Three Vertical Zones



Stretch-level



Eye-level



Stoop-level



## Stretch level — top shelf ( $\approx 72''$ )

Above the line of sight; shoppers must **look and reach up**.

## Eye level — second shelf ( $\approx 60''$ )

Aligned with the average adult's **line of sight**; the first place the eye lands.

## Stoop level — fifth shelf ( $\approx 24''$ )

Below the line of sight; shoppers must **bend down**.

## Conventional wisdom

Eye-tracking: attention is highest at **eye level**, then **stretch**, then **stoop**. Hence “eye level is buy level” — and why brands pay for it.

# Industry Partner: A Field Experiment in Convenience Stores

## Our partner

One of the largest **North American convenience-store chains** (thousands of stores, multiple countries); 1,500 SKUs in 35 categories.

## The 2022 experiment

- Randomly selected stores; **2 treatment + 3 control**.
- **14 fridges** per store (juices, water, energy drinks, beer, wine...): 38% of SKUs, **45% of sales**.
- Only **vertical position** changed — same products, facings, and horizontal spots.
- Field team of 5 RAs **photographed shelves daily** to ensure compliance.

## Why this is credible

Randomized stores, pre-set planograms, and daily monitoring  $\Rightarrow$  close to a **randomized field experiment**, not a correlation study.

## Rotation over time

Round 0: baseline (P0)  
Round 1: **Reorg-1** (swap)  
Round 2: **Reorg-2** (rotate)  
Rounds 3–4: robustness checks

# Same Eye-Level Spot, Very Different Lift

**Before  
(P0)**

**Reorg-1  
swap**

**Reorg-2  
rotate all**

*Store 1: A moves stretch → eye*

|         |          |
|---------|----------|
| Stretch | <b>A</b> |
| Eye     | B        |
| Stoop   | C        |

|          |
|----------|
| B        |
| <b>A</b> |
| C        |

**+5.6%**

|          |
|----------|
| C        |
| <b>A</b> |
| B        |

**-0.5% n.s.**

*Store 2: C moves stoop → eye*

|         |          |
|---------|----------|
| Stretch | A        |
| Eye     | B        |
| Stoop   | <b>C</b> |

|          |
|----------|
| A        |
| <b>C</b> |
| B        |

**+10.0%**

|          |
|----------|
| B        |
| <b>C</b> |
| A        |

**+2.0% n.s.**

Badges: sales lift of the moved product vs. control stores.

- Same product, **same eye-level spot** — different neighbors.
- **Swap**: +5.6% (from stretch), +10% (from stoop).
- **Rotate all**: no significant lift.
- On average eye level **does** sell more (+6.9% vs. stoop, +4.4% vs. stretch) — the average **hides** the reorganization.
- Confusion, order, and round length ruled out.

# Why? Partial Visual Search

## Minimal searchers

look at eye level only  
choice set: {A} in both

## Partial searchers

eye level + stretch level  
{A,B} vs. {A,C}

## Full searchers

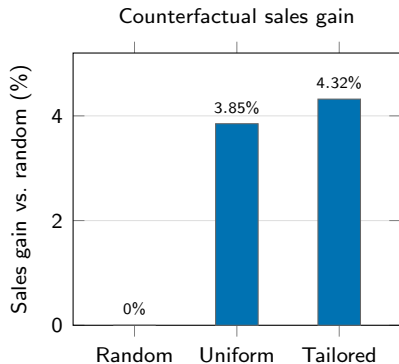
scan every shelf  
choice set:  
{A,B,C} in both

- Shoppers scan **eye level first**, then the shelf **above**, then below — and stop when satisfied.
- Only partial searchers see a different **competitor** next to A. If that neighbor is **strong**, A's eye-level advantage disappears.
- Confirmed statistically: the effect runs **entirely** through how concentrated preferences are between eye and stretch level.

## Takeaway

Eye level is buy level — **when the product next to you is weaker.**

# The Money: Tailor the Reorganization by Shelving Unit



- Reorg-1 lifted **total fridge sales** by 3.6–3.8%; Reorg-2 by nothing.
- Yet Reorg-2 was **better** in 21% (stretch) and 14% (stoop) of fridges.
- **Uniform** = Reorg-1 in every fridge;  
**Tailored** = best reorganization per fridge — another half point.

## Contract implications

Slotting fees should price the **whole planogram**, not just “eye level”: exclusivity or category captaincy protects the lift; retailers can charge more when a contract forces a sub-optimal reorganization.

# Context: Action Items

## For retailers and brands

- 1 Before promising eye level, check **who ends up one shelf above**.
- 2 Decide the reorganization **fridge by fridge**, not chain-wide.
- 3 Negotiate a **menu of contracts**: “eye level + weak neighbor” is worth more than “eye level.”
- 4 Measure with **control stores** — averages across stores mislead.

## AI Where AI fits

- **Computer vision** on daily shelf photos (as our field team did manually) for compliance and sales attribution.
- Planogram optimizers should model **neighbor effects** and search depth, not position alone.
- Same logic online: rank 1 in search results pays off **only if rank 2 is weaker**.

## 4. Fairness: Size-Inclusive Fashion Assortments

### Why we studied it

- Top-10 UK womenswear: size 20 available **53%** of the time vs. **76%** for size 10.
- **67%** of American women wear size 14+.
- D&I perceptions affect purchase decisions of **80%** of customers.
- Revenue-maximizing algorithms **ignore** who is left out.

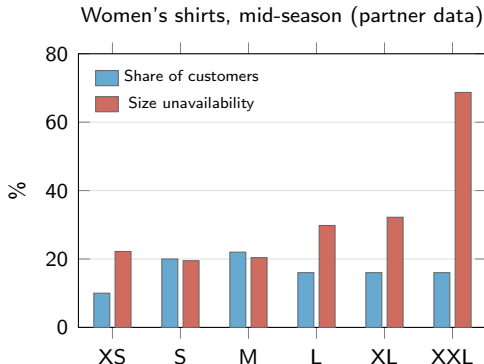
### Our partner

One of **Europe's largest online fashion retailers**:  $\approx 30,000$  options per season;  $>75\%$  of SKUs are **never replenished** in-season.

### Managerial question

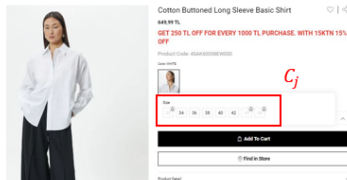
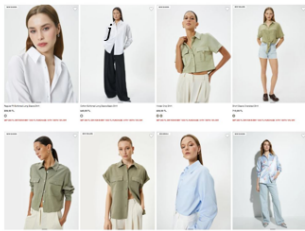
Inventory is fixed mid-season. Which products should we **display** to maximize revenue while keeping size groups' experiences **fair**?

# The Problem in One Chart



- L and XXL: **same customer share**; XXL is unavailable **2×** as often.
- The partner raised XXL from **0.2%** to **28%** of SKUs — the gap **persisted**.
- No one intended this: it emerges from ordinary **operational decisions**.

# How Customers Shop Online: Option First, Size Second



## Step 1: Landing (category) page

The shopper sees the **displayed options** (style × color) and clicks one. **No size information** is shown yet. Chance of clicking an option = its appeal relative to everything displayed (plus leaving).

## Step 2: Option page

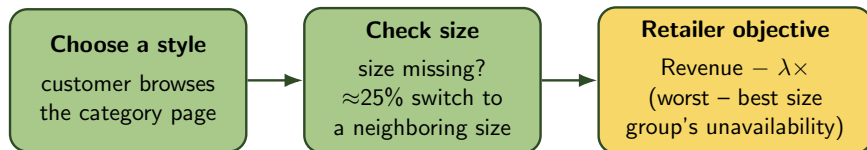
Only now does she see the **size configuration**. She buys her size, **switches to a neighboring size** ( $\approx 25\%$ ), or **leaves**.

# Why This Journey Creates a Fairness Problem



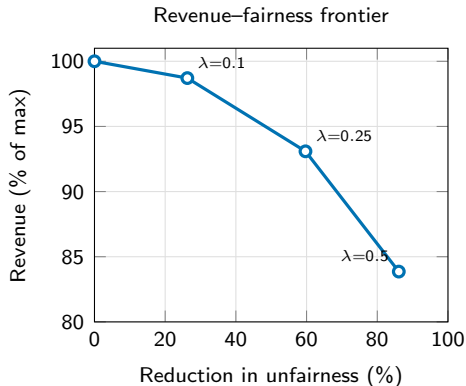
- The landing page is **size-blind**, but the outcome is **size-specific**: the disappointment arrives *after* the click.
- Fairness measure: each size group's **experienced unavailability** — how often a click leads to a missing size. Unfairness = worst-served minus best-served group.
- Mid-season, the retailer cannot restock — but it **controls which options appear on the landing page**. That is the lever.

# Our Approach: Fairness-Aware Display



- $\lambda$  = the **price of unfairness** the retailer is willing to pay.
- The retailer **rotates** among a few assortments (online, different visitors can see different pages).
- Solved exactly and fast; uses at most  $(\# \text{sizes})^2$  assortments — **implementable**.
- Key fact: if every product had every size, the revenue-optimal display would **already be fair**. Unfairness comes from **broken size runs**.

# Fairness Is Cheap at First, Expensive at the End



- **–26% unfairness** for only **1.3%** of revenue.
- **–60% unfairness** for **6.9%** of revenue.
- **–86% unfairness** costs **16.1%** — and starts making **well-served sizes worse** (“leveling down”).
- Implemented by **pruning** weaker options and **light rotation** — most of the commercial assortment stays.

# Operational Levers Change the Trade-Off

| At $\lambda = 0.1$      | Revenue | Unfairness reduced | How it works          |
|-------------------------|---------|--------------------|-----------------------|
| Baseline display        | 98.7    | 26%                | —                     |
| Size filter first       | 109.2   | 55%                | prevents bad exposure |
| Revisit after stock-out | 105.1   | 34%                | recovers lost sales   |

Revenue indexed to 100 = revenue-maximizing display without fairness.

## Win-win

Asking customers for their size **before** showing products raises revenue **and** fairness: personalization used for inclusion.

## No universal $\lambda$

The same fairness weight has different effects across categories. Each retailer should map **its own frontier** per category.

# Fairness: Action Items

## For merchandising leaders

- 1 Track a **service KPI by customer group** (e.g., unavailability by size), not just revenue.
- 2 Capture the **cheap first 25%** of fairness now.
- 3 Make **size filtering** the default entry point.
- 4 Fix upstream too: size curves estimated from censored sales **understate** plus-size demand.

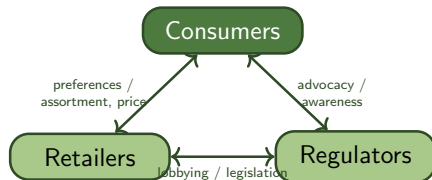
## AI Where AI fits

- Recommenders and ranking engines inherit inventory biases — **audit outcomes by group**.
- Add fairness as an **explicit term** in the objective, with a board-approved  $\lambda$ .
- Correct demand models for **stock-out censoring** before using them to buy.

## 5. Green: Profit vs. Environmental Impact in Fast Fashion

### Why we studied it

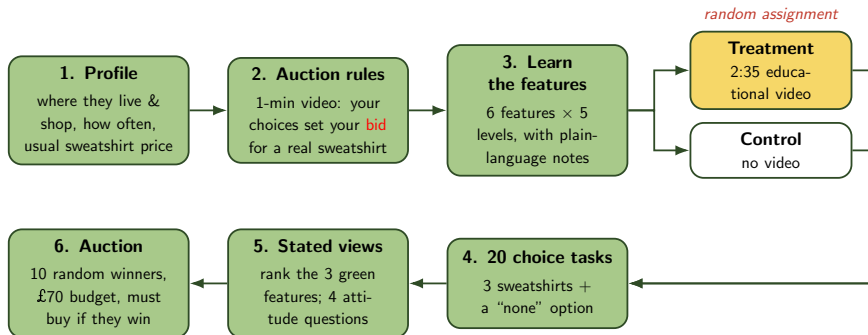
- **Say-do gap**: surveys show willingness to pay; retailers say customers won't.
- Low WTP is cited as the **#1 barrier**; weak regulation **#2**; low awareness **#4**.
- Regulators crack down on **greenwashing** (e.g., EU-Zalando labels; Norway-H&M claims).



### Managerial questions

What will customers **really** pay for green attributes? What does a green mandate **cost** a retailer? Can **awareness** soften the trade-off?

# The Experiment: A Respondent's Journey



## Sample

**489 UK shoppers** (Prolific), ages 18–45, recently bought from a fast-fashion retailer; **231 treatment, 258 control**; ≈19 minutes each.

## Analysis pipeline

Individual preferences → **WTP** per feature → optimize a 5-product line & prices under green targets.

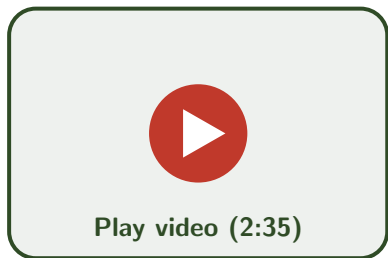
# What Respondents Saw: Six Features, Five Levels Each

| Feature                   | Levels                                              | How it was explained                                                                         |
|---------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------|
| Bill of materials         | 100% cotton to 100% polyester (5 blends)            | composition only — <i>not</i> labelled as greener                                            |
| Type                      | stylish, lounge, casual, active, essential          | sweatshirt sub-category                                                                      |
| Price                     | £10, £25, £40, £55, £70                             | fast-fashion range plus a green premium                                                      |
| <b>Recycled content</b>   | 0, 15, 30, 45, 60%                                  | share of <i>both</i> fibres that is recycled                                                 |
| <b>Durability</b>         | none, +10, +20, +30, +40% lifespan                  | “worn 100 times → 130 times” at +30%                                                         |
| <b>Circularity design</b> | none, repairs, redistribution, upcycling, recycling | warranty/DIY repair; take-back & resale; remake into new items; avoid hard-to-recycle blends |

## Design choice

Many green levels **are not on the market yet**, so only an experiment can price them — and each has **several levels**, not just yes/no.

# The Treatment: A Short Educational Video



## What it does

- Introduces each green feature and **how higher levels reduce** a garment's environmental impact.
- Framed neutrally: "the retailer wanted to share additional information."
- Shown to a **random half** of respondents, *before* the choice tasks.

## Question it answers

Can a **2.5-minute** awareness nudge change what people choose and pay?

## As you watch

You are now the **treatment group**. On the next slide, make a choice and ask yourself whether the video changed it.

# Your Turn: One of the 20 Choice Tasks

If these were your only options, which would you choose?

(1 of 20)

|                                   |                                       |                                       |                                       |                                                                                                        |
|-----------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------|
| Bill of materials                 | 100% polyester                        | 50% cotton, 50% polyester             | 50% cotton, 50% polyester             | <b>NONE:</b> I would rather keep my money, or purchase a sweatshirt from a competing fashion retailer. |
| Functionality                     | Casual                                | Stylish                               | Lounge                                |                                                                                                        |
| Recycled content                  | 0% recycled content                   | 0% recycled content                   | 30% recycled content                  |                                                                                                        |
| Durability (increase in lifespan) | 10% increase                          | 30% increase                          | None                                  |                                                                                                        |
| Circularity design                | Designed for redistribution           | Designed for recycling                | Designed for repairs                  |                                                                                                        |
| Price (£)                         | 25                                    | 40                                    | 25                                    |                                                                                                        |
|                                   | <input type="button" value="Select"/> | <input type="button" value="Select"/> | <input type="button" value="Select"/> | <input type="button" value="Select"/>                                                                  |

Which would you choose?

“None” = keep your money, or buy from a competitor.

- **15 personalized** tasks: each person sees a different, balanced design that forces **hard trade-offs**.
- **5 identical** validation tasks (every 4th screen) test the model.
- Choices reveal trade-offs that **ratings hide**: recycled content here means giving up durability.

# Making Choices Count: The Auction

## Incentive alignment

- 10 respondents drawn at random enter an auction for a real sweatshirt, with a £70 budget.
- Their bid is computed from their choices — nobody states a price.
- Win  $\Rightarrow$  must buy at the auction price.
- Understate WTP  $\rightarrow$  lose the item; overstate  $\rightarrow$  overpay. Truth-telling is the best strategy.

## Why it matters

Hypothetical surveys overstate willingness to pay for “green” — the say-do gap this study targets.

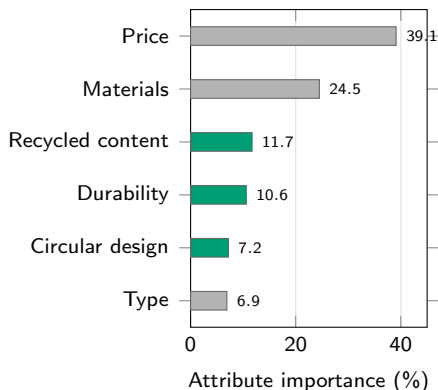
## The auction sweatshirt

|                  |               |
|------------------|---------------|
| Materials        | 100% cotton   |
| Type             | Casual        |
| Recycled content | 50%           |
| Durability       | +10% lifespan |
| Circularity      | none          |

## Did the model work?

On the 5 held-out tasks it predicts the actual choice 59.5% of the time — vs. 25% by chance (4 options).

# What Customers Actually Value



- Price rules — but green attributes together are  $\approx 30\%$  of the decision.
- **Stated vs. revealed:** 63% say durability matters most; choices favor **recycled content**.
- **Step function:** any green level beats none; higher levels add less utility.
- $\approx 43\%$  are environmentally leaning (two segments) — often frequent shoppers.

# Green: Action Items

## For retailers and policymakers

- 1 Offer the **first level** of each green attribute broadly — that is where most value lies.
- 2 Keep a **mixed** product line; uniform mandates on every item are costly.
- 3 **Sequence** mandates gradually; order matters for the burden.
- 4 Invest in **education**, not just labels; trust revealed, not stated, preferences.

## AI Where AI fits

- Choice models + optimization = a **“digital twin” of the customer** to test green lines before launch.
- Validate AI-generated “synthetic respondents” against **incentive-aligned** data — stated answers misled here.
- Use GenAI to **personalize education**, with claims checked to avoid greenwashing.

# Three Lessons That Cut Across All Five Studies

## 1. Context matters

Eye-level lift depends on the neighbor; threshold response depends on recency; 4 green segments.

⇒ **tailor**

## 2. Data reflects a static story

Observed cart value includes top-up; sales history is censored by stock-outs.

⇒ **experiment**

## 3. Say $\neq$ do

Stated green priorities and survey complaints diverge from behavior.

⇒ **measure revealed choices**

## For AI projects

Train on **experiments and natural experiments**, segment by **behavioral mechanism**, and roll out **gradually** with control groups.

# The AI Playbook for Human Responses

| Lens      | Human response                       | Data signal                           | AI application                                       |
|-----------|--------------------------------------|---------------------------------------|------------------------------------------------------|
| Framing   | zero-price effect; fee aversion      | S&H-to-price ratio, competitor prices | choose the <b>price frame</b> per SKU                |
| Reference | past thresholds anchor goals         | recency, cart-completion time         | <b>personalized rollout</b> ; top-up recommendations |
| Context   | partial visual search                | shelf photos, neighbor sales          | <b>vision-based</b> planogram & ranking optimization |
| Fairness  | unequal service by group             | unavailability by size                | <b>fairness term</b> in assortment/ranking objective |
| Green     | step-shaped, awareness-sensitive WTP | incentive-aligned choices             | <b>digital-twin</b> product-line design              |

# Thank you.

Questions and discussion.

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